Task 6: Data Visualization Mastery with Seaborn

Day 1 : Seaborn Basic

importing libraries

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import plotly.express as px
import plotly.graph_objects as go

sns.set(style="whitegrid")
```

loading dataset

```
df = pd.read_csv('/content/sales_data (4).csv')
df.head()
```

	Date	Product	Quantity	Price	Customer_ID	Region	Total_Sales
0	2024-01-01	Phone	7	37300	CUST001	East	261100
1	2024-01-02	Headphones	4	15406	CUST002	North	61624
2	2024-01-03	Phone	2	21746	CUST003	West	43492
3	2024-01-04	Headphones	1	30895	CUST004	East	30895
4	2024-01-05	Laptop	8	39835	CUST005	North	318680

```
#Basic info
df.info()
df.describe()
```

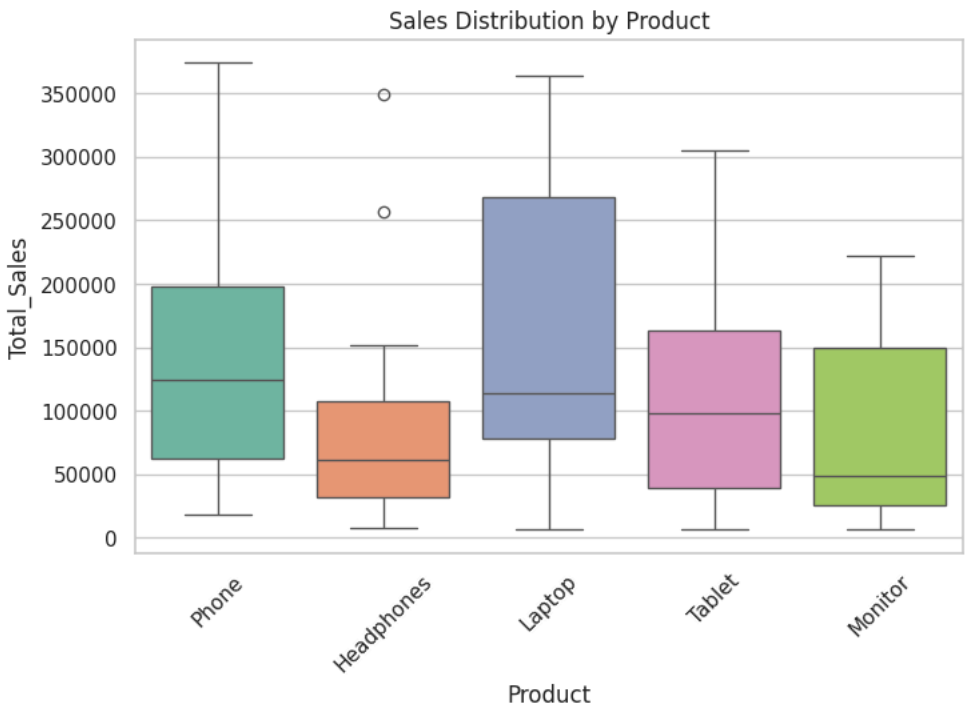
```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 100 entries, 0 to 99
Data columns (total 7 columns):
 #   Column       Non-Null Count  Dtype
---  ------       --------------  -----
 0   Date         100 non-null    object
 1   Product      100 non-null    object
 2   Quantity     100 non-null    int64
 3   Price        100 non-null    int64
 4   Customer_ID  100 non-null    object
 5   Region       100 non-null    object
 6   Total_Sales  100 non-null    int64
dtypes: int64(3), object(4)
memory usage: 5.6+ KB
```

	Quantity	Price	Total_Sales
count	100.000000	100.000000	100.000000
mean	4.780000	25808.510000	123650.480000
std	2.588163	13917.630242	100161.085275
min	1.000000	1308.000000	6540.000000
25%	2.750000	14965.250000	39517.500000
50%	5.000000	24192.000000	97955.500000
75%	7.000000	38682.250000	175792.500000
max	9.000000	49930.000000	373932.000000

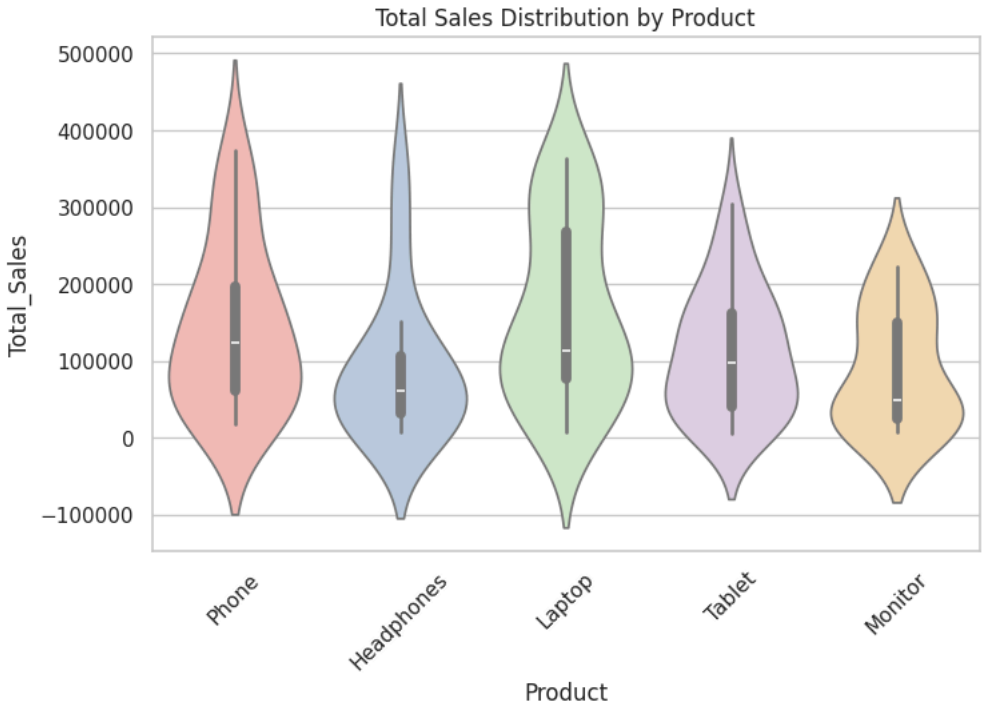
```
#converting data column
df['Date'] = pd.to_datetime(df['Date'])
df['Year'] = df['Date'].dt.year
df['Month'] = df['Date'].dt.month
```

Day2: Statistical Visulizations

```
#box plot
plt.figure(figsize=(8,5))
sns.boxplot(x='Product' , y='Total_Sales', data=df, palette='Set2', hue='Product', legend=False)
plt.title('Sales Distribution by Product')
plt.xticks(rotation=45)
plt.show()
```




```
#violin plot
plt.figure(figsize=(8,5))
sns.violinplot(x='Product', y='Total_Sales' , data=df, palette='Pastel1', hue='Product', legend=False)
plt.title('Total Sales Distribution by Product')
plt.xticks(rotation=45)
plt.show()
```

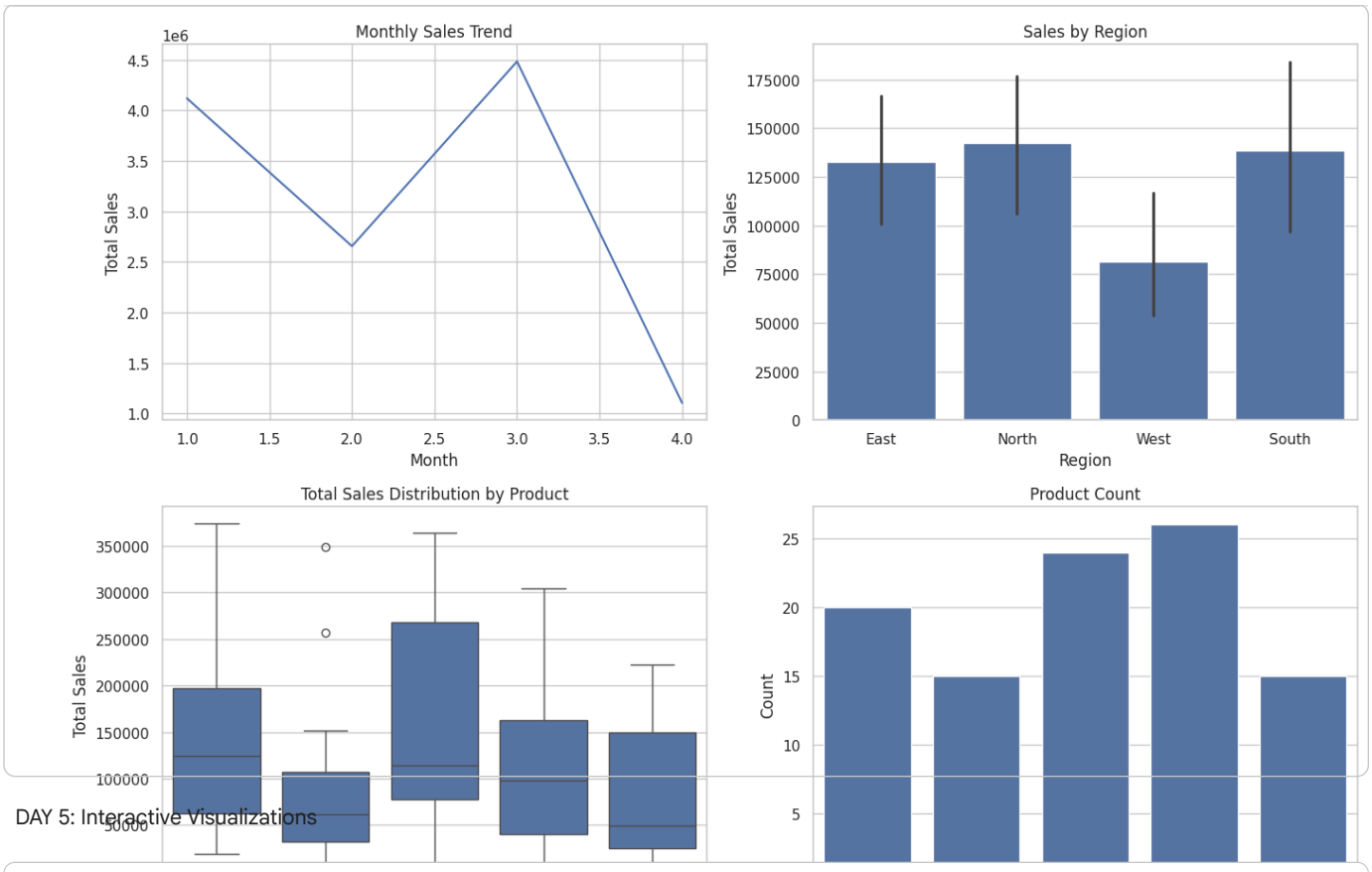


Heatmap & Correlation

```
#correlation Matrix
plt.figure(figsize=(10,6))
corr = df.corr(numeric_only=True)

sns.heatmap(corr, annot=True , cmap="coolwarm", fmt=".2f")
plt.title("correlation heatmap")
plt.show()
```

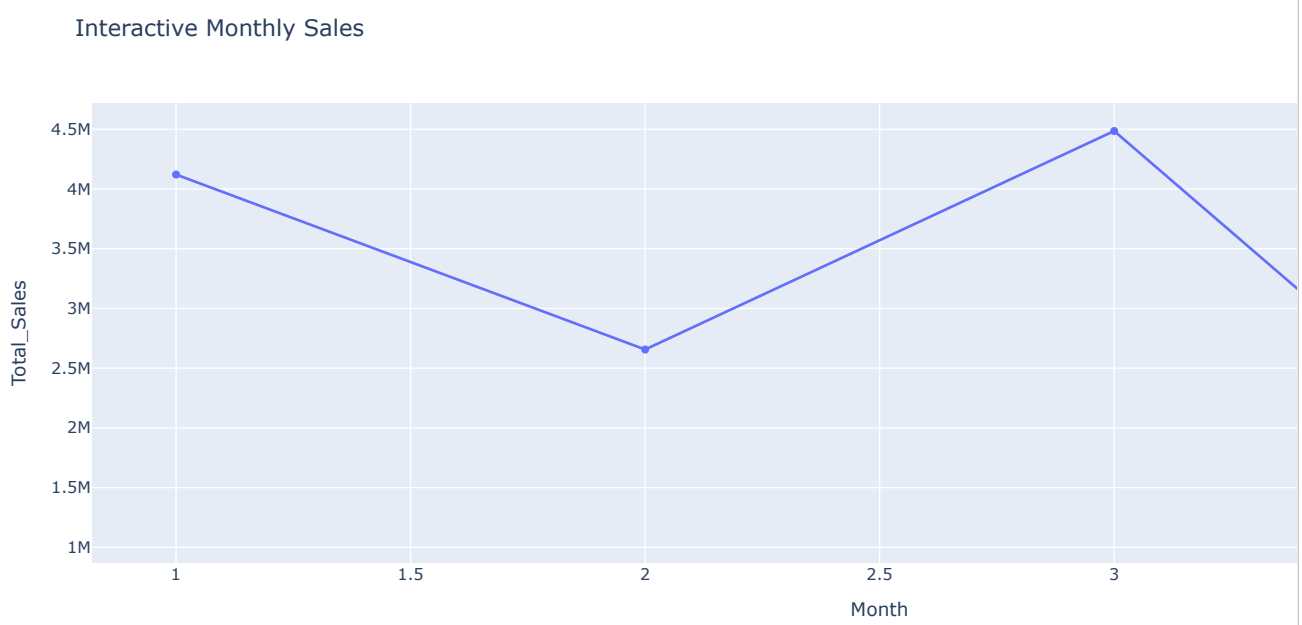
DAY 4 — Multi-Plot Dashboard

```
fig, axes = plt.subplots(2,2, figsize=(14,10))

#sales trend
monthly_sales = df.groupby('Month')['Total_Sales'].sum()
axes[0,0].plot(monthly_sales)
axes[0,0].set_title('Monthly Sales Trend')
axes[0,0].set_xlabel('Month')
axes[0,0].set_ylabel('Total Sales')

#sales by reagion
sns.barplot(x='Region', y='Total_Sales', data=df, ax=axes[0,1])
axes[0,1].set_title('Sales by Region')
axes[0,1].set_xlabel('Region')
axes[0,1].set_ylabel('Total Sales') # Added y-label for clarity

#profit by category (replaced with Total Sales by Product)
sns.boxplot(x='Product', y='Total_Sales', data=df, ax=axes[1,0])
axes[1,0].set_title('Total Sales Distribution by Product')
axes[1,0].set_xlabel('Product') # Added x-label
axes[1,0].set_ylabel('Total Sales') # Added y-label

# Segment Count (replaced with Product Count)
sns.countplot(x='Product', data=df, ax=axes[1,1])
axes[1,1].set_title('Product Count')
axes[1,1].set_xlabel('Product') # Added x-label
axes[1,1].set_ylabel('Count') # Added y-label

plt.tight_layout()
plt.show()
```

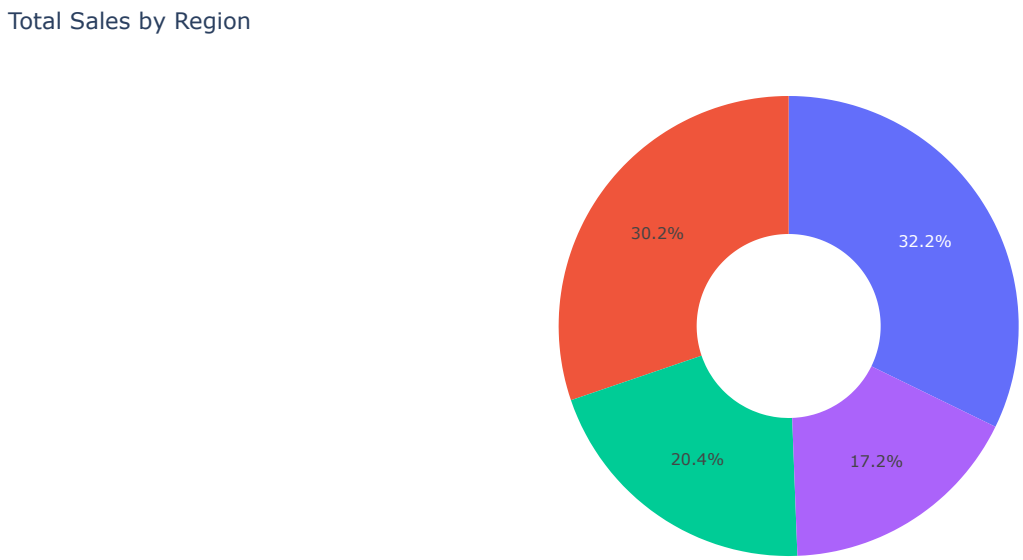


DAY 5: Interactive Visualizations

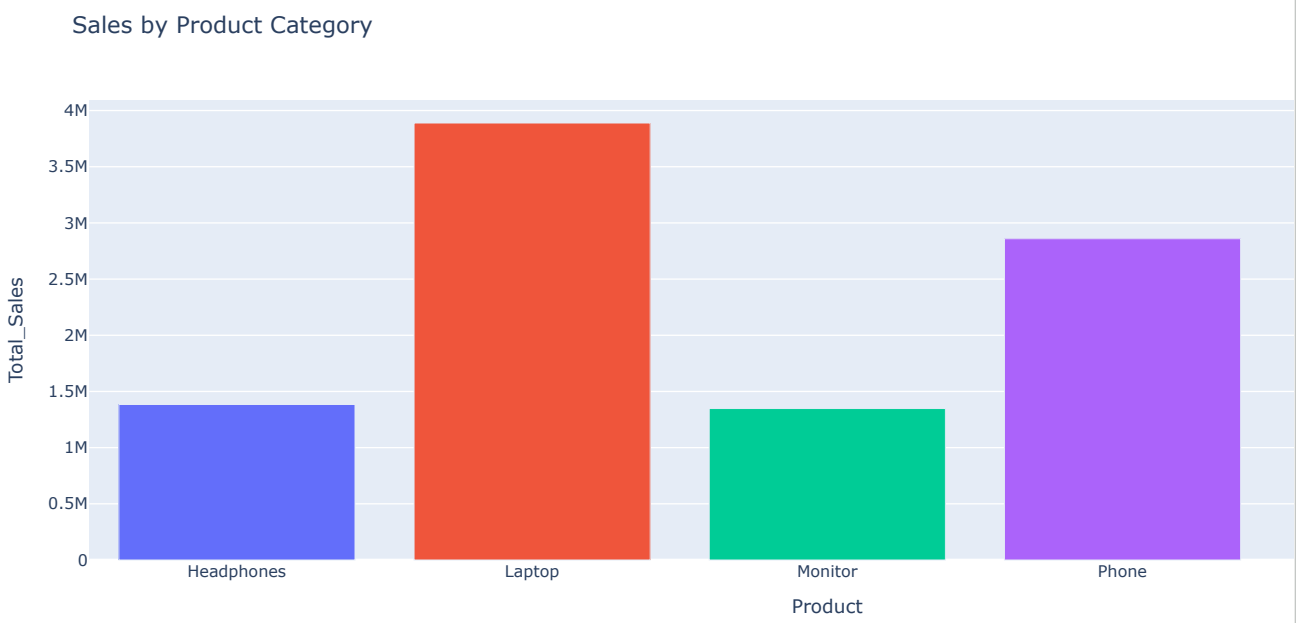
```
#Interactive Sales Trend
fig = px.line(df.groupby('Month')['Total_Sales'].sum().reset_index(),
              x='Month', y='Total_Sales',
              title='Interactive Monthly Sales',
              markers=True)

fig.show()
```




```
#Interactive Customer Segmentation Pie. (Changed to Sales by Region)
fig = px.pie(df, names='Region', values='Total_Sales',
             title='Total Sales by Region',
             hole=0.4)
fig.show()
```



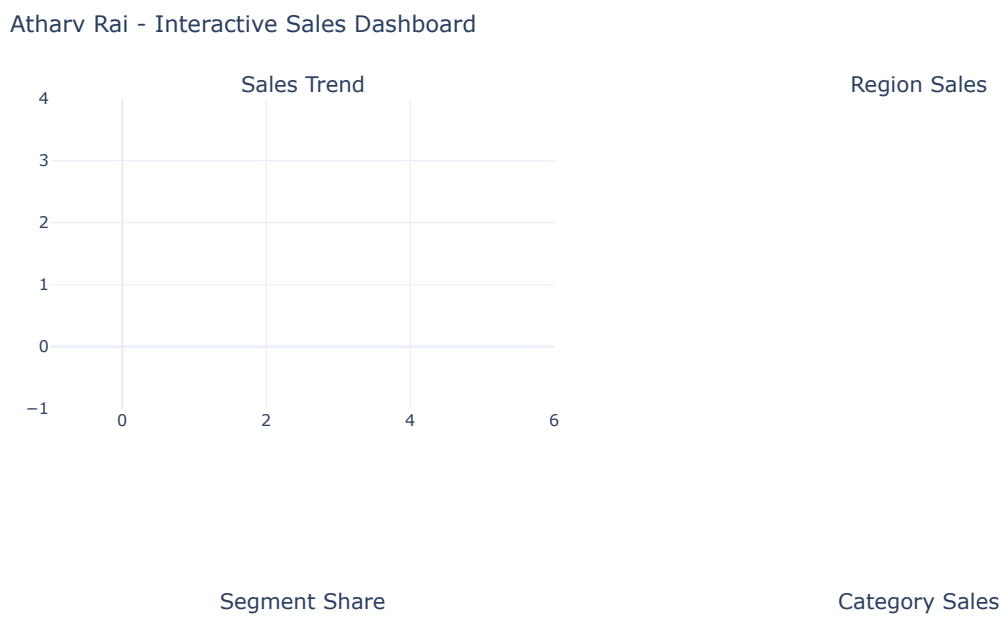

```
#Interactive Product Performance.
fig = px.bar(df.groupby('Product')['Total_Sales'].sum().reset_index(),
             x='Product', y='Total_Sales',
             color='Product',
             title='Sales by Product Category')

fig.show()
```



DAY 6: Dashboard Integration

```
from plotly.subplots import make_subplots

fig = make_subplots(
    rows=2, cols=2,
    subplot_titles=("Sales Trend","Region Sales","Segment Share","Category Sales")
)
```

```
# Add traces accordingly

fig.update_layout(
    height=800,
    width=1000,
    title_text="Atharv Rai - Interactive Sales Dashboard",
    template="plotly_white"
)

fig.show()
```



DAY 7: Polish & Presentation

```
#Cohesive Color Theme
sns.set_theme(style="whitegrid", palette="Set2")
```

```
#Add Branding
plt.suptitle("Atharv Rai - Sales Dashboard Project", fontsize=16)
```

```
Text(0.5, 0.98, 'Atharv Rai - Sales Dashboard Project')
<Figure size 640x480 with 0 Axes>
```